HCML Monitoring Portal: Server flow (one request through the BFF)

Browser
nginx portal-web
Fastify portal-bff
TTL cache in memory
Zabbix API JSON-RPC 2.0
Mail server SMTP, optional
portal-state volume
1
GET /bff/api/sites
Cookie: hcml_session=<jwt>
2
GET /api/sites
Host with its port, X-Forwarded-For preserved
The plugin chain, in registration order
3
helmet: security headers
CSP off: this process serves JSON only
4
CORS: one allowed origin (WEB_ORIGIN)
5
rate limit: per IP, before auth,
so a flood is rejected before JWT work
6
compression hook: brotli / gzip
alt
[rate limit exceeded]
7
429 rate_limited
"the limit is N per minute"
onRequest: forgery check, then the auth guard
8
forgery check, unsafe methods only: a cookie POST
needs X-HCML-CSRF: 1 and an Origin naming this host
9
verify the JWT: Bearer header, else the cookie.
Never ?token=, not even on the stream
10
requiredRole(url): first match wins,
anything unmatched needs viewer
alt
[cookie POST without the header, or from another site]
11
403 csrf, before any token is checked
[no or bad token]
12
401 unauthorized: "Sign in to continue."
[role too low]
13
403 forbidden: names the role needed
14
cached("sites", 15 s, fn)
alt
[fresh, within TTL]
15
the stored value
no Zabbix call at all
[another request is already fetching]
16
await the same promise
coalesced: twenty NOC screens
cost one upstream read
[stale but inside staleMs]
17
the old value, now
18
one background refresh
a failed refresh keeps the old value
and is logged, not thrown
[miss]
19
host.get · problem.get · trigger.get · item.get
20
raw JSON-RPC result
21
store, LRU, max 500 keys
The part Zabbix cannot do
22
resolveSite(): tag, then host name,
then inventory, then group
23
roll hosts up per site, worst severity wins
24
200 application/json
content-encoding: br
25
same, unbuffered
When Zabbix does not cooperate
26
token rejected → 503 zabbix_auth
"fix the token", not an opaque 500
27
no answer in ZABBIX_TIMEOUT_MS → 503 zabbix_timeout
28
JSON-RPC error → 502 zabbix_error, carrying Zabbix's own words
the address in ZBX_URL goes to the log,
never into the response body
The one write to Zabbix, for comparison
29
POST /api/problems/acknowledge, operator only
the cookie plus X-HCML-CSRF: 1
30
event.acknowledge, on the SEPARATE write token
31
invalidate("problems", holdMs)
32
push the new list to every open screen at once
The report by email: administrators only, and off until MAIL_ is set
33
POST /api/sli/email
the cookie plus X-HCML-CSRF: 1
34
month, files and addresses checked: one to ten,
inside MAIL_ALLOWED_DOMAINS, five sends an hour each
35
log in first, so a wrong password
fails in a second
36
the month's SLA, both methods
from the cache, or from Zabbix on a miss
37
one message: the figures in the text,
the files attached
38
the record of the send: counts and a word,
never an address
39
how many addresses the mail server took
On the chosen day the scheduler does the same without a browser,
and saves each step to the state file before the next
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